

Blues Deluxe mod



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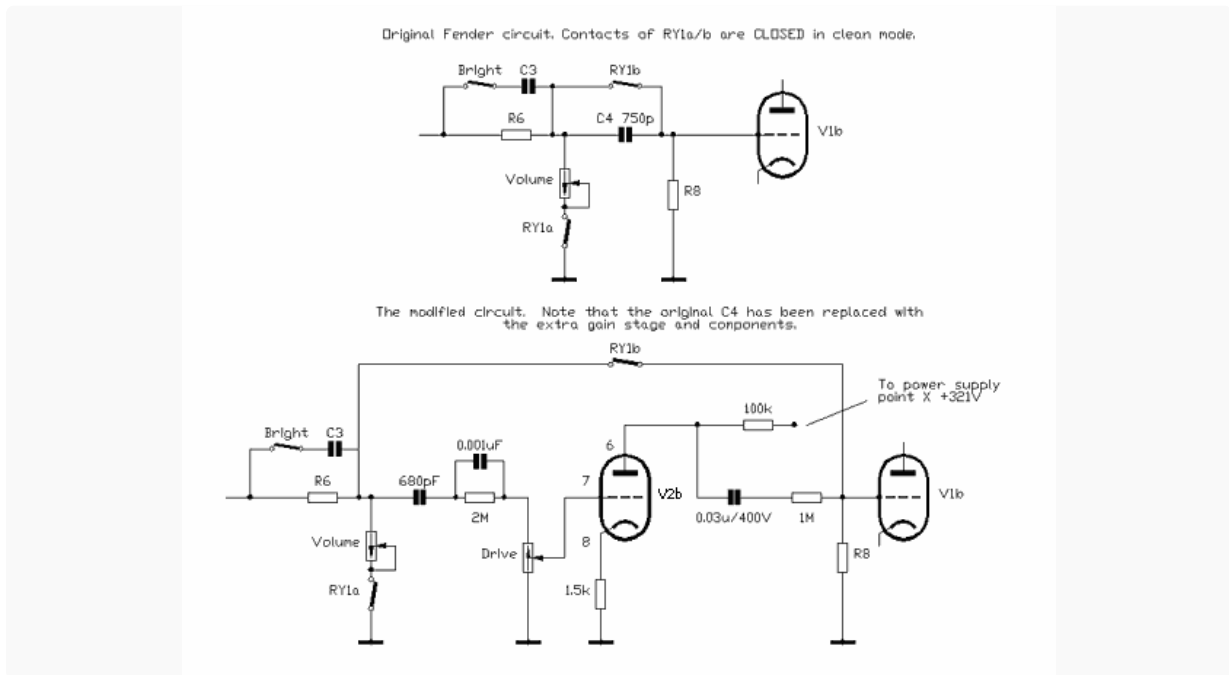
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Andy Marshman kindly let me have the following notes which described how to liven up an unused triode section in the Fender Blues Deluxe in order to give more gain. Main features are as follows:

- No physical changes to the outside of the amplifier.
- No changes of potentiometer values.
- No need to buy or install extra tubes or sockets.
- Clean channel maintains its TRUE BYPASS function.
- Amp user can use the Drive and Master pots in a conventional manner, i.e. where the distortion is controlled by the Drive control and the overall loudness by the Master. In Fenders design, the Drive channel is just a louder, still clean version of the Clean channel.

Here are Andy's notes - please read them carefully, and please don't attack your amp unless you are 100% sure of what you are doing!

Disclaimer: Prior to starting, disconnect power cord and ensure all filter capacitors are fully discharged. Use the services of a trained technician if you are unsure about any portion of this modification. This modification will void your warranty. Study and understand the entire procedure before starting. Modifying a printed circuit board amplifier is more difficult and unconventional than an eyelet point to point board. Some unusual connection techniques are required, e.g. soldering a wire to the bottom of a tube socket connection. The author assumes no responsibility or liability for equipment damage or personal harm from individuals working on their own amplifier.



Initial Preparations

Note: by step 5, you will have to remove the chassis from the cabinet. Optionally, it can be removed right at the beginning.

1Cut tracks on socket of V2B that ground pins 6,7,8. Ensure none of these three pins are grounded.**2**Solder 3 flying leads (16" long for starters) to pins 6,7,8. Different colours would be a good idea. Secure them to a fixed point, typically above at the main PCB, with tie-wraps to ensure no stress is put on solder joints at the tube socket.**3**Examine the two partial schematics of the original circuit and the modified one. Note that cap C4 is being cut out and replaced with the extra tube stage V2B. The relay contacts of RY1A and 1B are closed in Clean mode, thereby giving true Bypass function around the extra gain stage.**4**The cap 680pF could be substituted with the original C4 of 750pF if desired. This cap is connected to R6, C3 on the incoming side. The other side of this cap goes directly up to the newly mounted Auxilliary Circuit Board, shown below. The shorter the leads, the less chance of picking up extraneous noise.**5**A significant change has to be made to the connections at the Drive potentiometer. Trace the PCB tracks to a couple of wire jumper links leading away from the Drive pot. Cut these out and do not substitute another resistor in place of the Drive pot. This takes the drive pot off the grid of V1b.

NB cut the PCB trace, right at the Drive pot that jumpers its wiper to the ground end. Leave the trace from ground intact going to the end of the pot. THIS STEP REQUIRES YOU TO REMOVE THE ENTIRE CHASSIS FROM THE CABINET.

At this time, solder a twisted pair of wires to the pot wiper and the top end of the pot. Again, about 16" long to start with. This will soon be trimmed shorter.

Solder the wire from the wiper of the Drive pot to pin 7 of V2B (grid).

Located the power supply filter capacitor C46. Follow the PCB trace over to R18. Tack solder a flying lead to this point. This will be the +ve supply to the new tube stage.

Located one of the chassis mounting screws and secure a solder lug under the screw with a flying lead (ground of course) for connection to the Auxilliary Board.

The Auxilliary Circuit Board

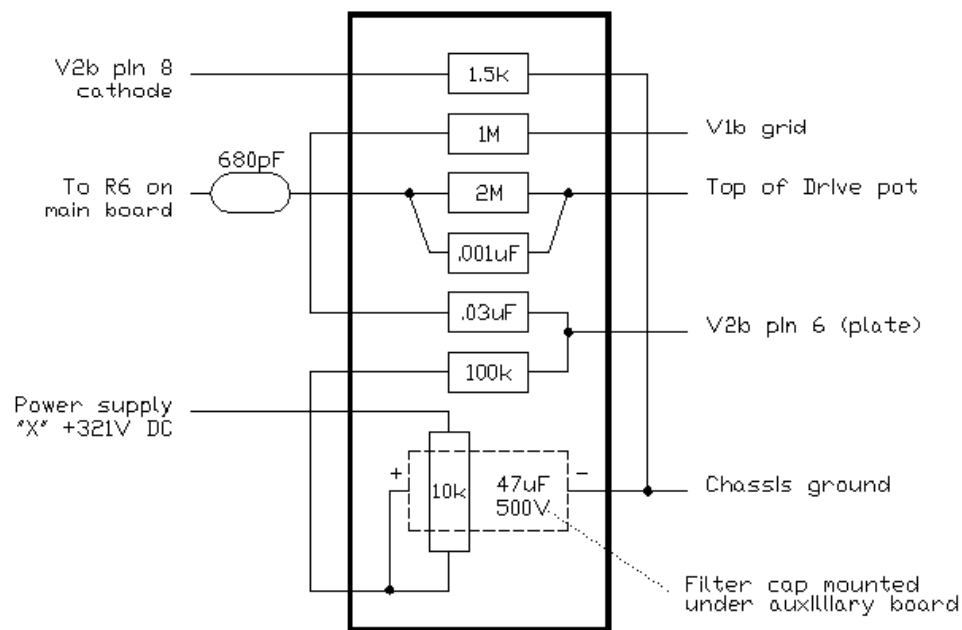
This is an added board that can be a breadboard, one with copper tracks on one side, or an eyelet board. It's mounted with standoffs so it clears the main PCB and allows the back cover to be installed and maintain clearance.

****** High voltage DC will be connected to this board in the completed circuit. Observe spacing clearances so that arcing does not occur between any adjacent components or copper tracks (if they exist on your board).

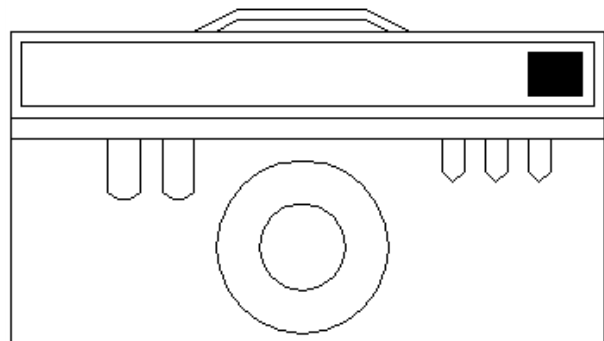
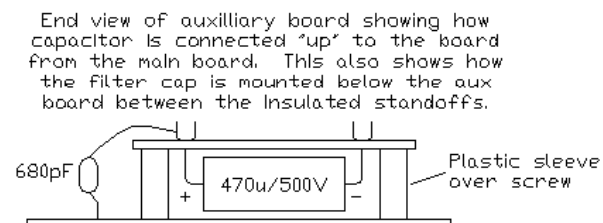
To mount the board, I used existing chassis screws on the main board; I removed them and replaced them with longer ones to hold up the Auxilliary Board. I also used plastic covers (from small Banana plugs) to cover the screws and act as a spacer as well. I was able to use one screw at each of the two diagonal corners.

Examine the diagrams, and note the location of the filter capacitor UNDER this board. This allows installation of the extra circuitry in a compact fashion, allowing clearance to the back cover of the amp. This back cover has a metallic foil on it for shielding. I added a piece of thin cardboard over that shielding in the vicinity of the Auxilliary Board to prevent high voltage jumping to the metal.

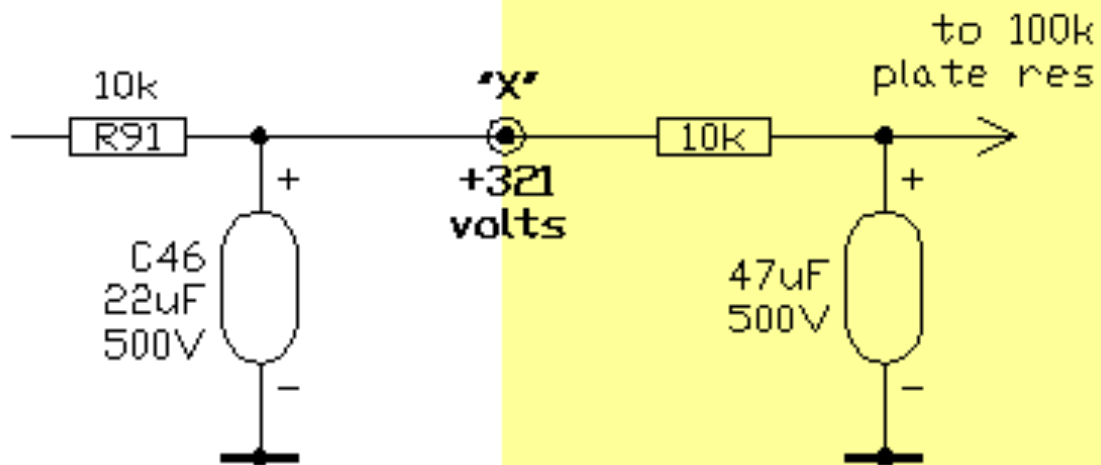
The layout sketch of the Auxilliary Board shows a typical component layout. This isn't critical. You may use your own point-to-point wiring techniques. Observe clearances for power supply voltages present on this circuit board.



All resistors are 1/2W 1%, the filter cap is rated at 500VDC, other caps are 400VDC.



Rear view of amplifier showing approximate location of auxilliary board



Connex new circuit to existing power supply at point "X"

Closing Notes

This modification adds tons of gain to the original circuit. You won't want to turn the Master all the way unless you're in a stadium. However, if you must have more gain than this provides, lower the values of resistors on the extra gain stage down from 1 meg, 2 meg to as low as 470k ohms. There is no capacitor across the cathode resistor for the same reason; adding one would only increase the stage gain. But again, experiment if you wish.

Fender designed a great amp. This mod just allows you to get a wide range of tones from subtle overdrive to high gain tube distortion when in the Drive mode.

Footnote:

Feedback from people who have done this mod indicates that the 47uF cap and 10K resistor are NOT ESSENTIAL and may be omitted. That means you don't need to build an auxiliary board, you may wire components directly to sockets and/or run lead wires from point to point to build the circuit.

Any questions or comments on the above can be emailed to me at agm1@rogers.com

Andy Marshman

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